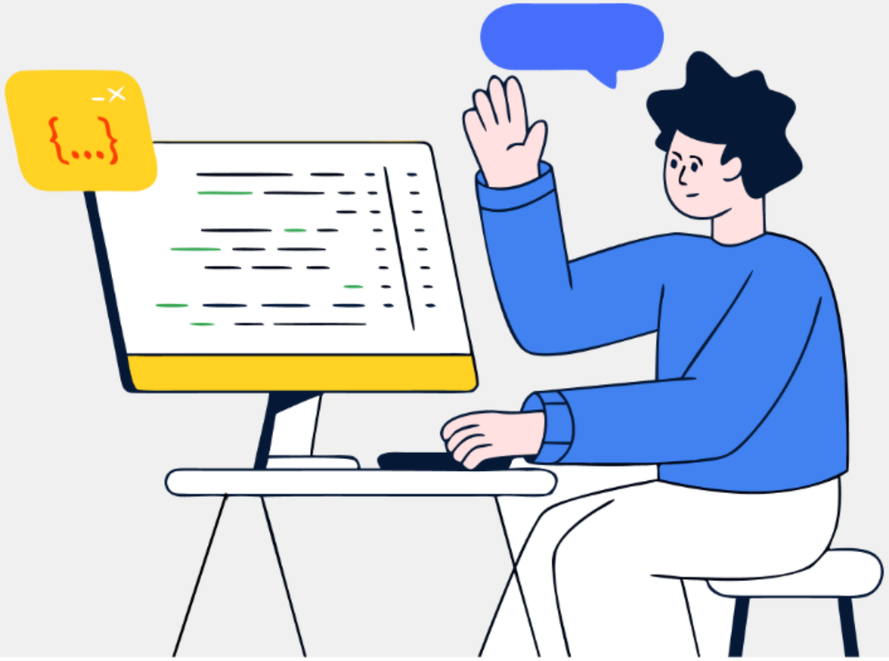
Build with AI

Solution Challenge



Team Details

a. **Team name: Docvail**

b. **Team leader name: Ankit Solanki**

c. **Problem Statement:**
- Many patients don’t know which doctor to consult or when they’re available.
- This leads to delays in diagnosis and treatment.
- Docvail solves this by making healthcare access faster and clearer.

Solution

- Docvail connects patients to the right doctor instantly.
- It acts as an intelligent bridge between patients and providers.
- First, it uses **Google Gemini 2.5 Flash** to analyze a patient's natural-language symptoms (including "Hinglish") to accurately determine the required medical specialist.
- Then, it instantly connects the patient to a live database of verified doctors in their city, showing real-time slot availability for immediate booking without the need for phone calls or waiting rooms.

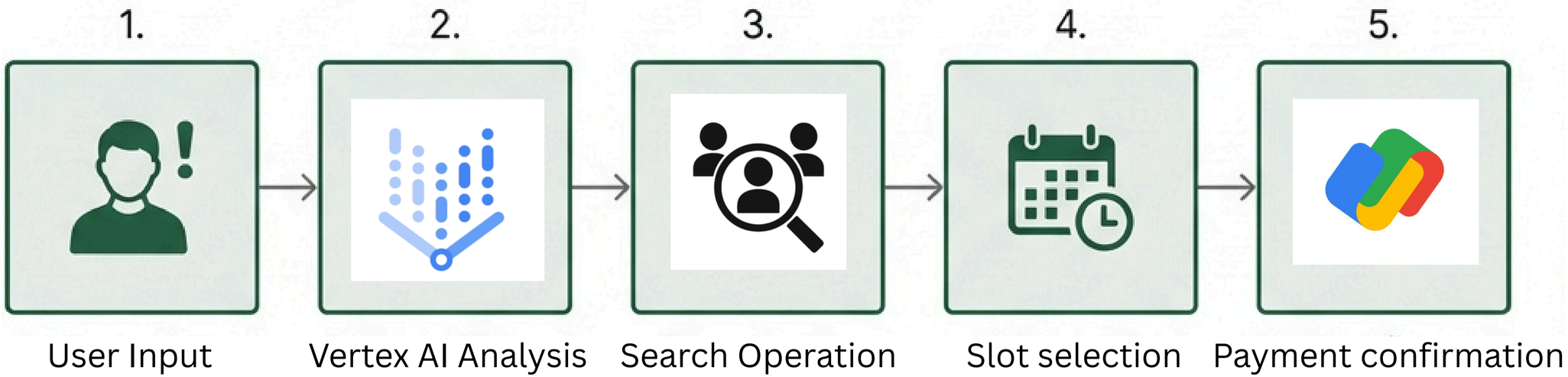
Opportunities

- By showing live OPD status, we eliminate wasted travel for rural patients and reduce hospital overcrowding by spreading patient arrivals throughout the day.
- Traditional platforms act as simple phonebooks (you have to know what kind of doctor you need) or symptom checkers (which only give advice, not an appointment).
- It eliminates the "knowledge gap" (AI tells you who to see) and the "access gap" (Live scheduling tells you exactly when they are free).
- The seamless integration of Google's state-of-the-art conversational AI (Gemini 2.5) with a secure, multi-tenant hospital scheduling architecture, accessible even to users typing in regional dialects.

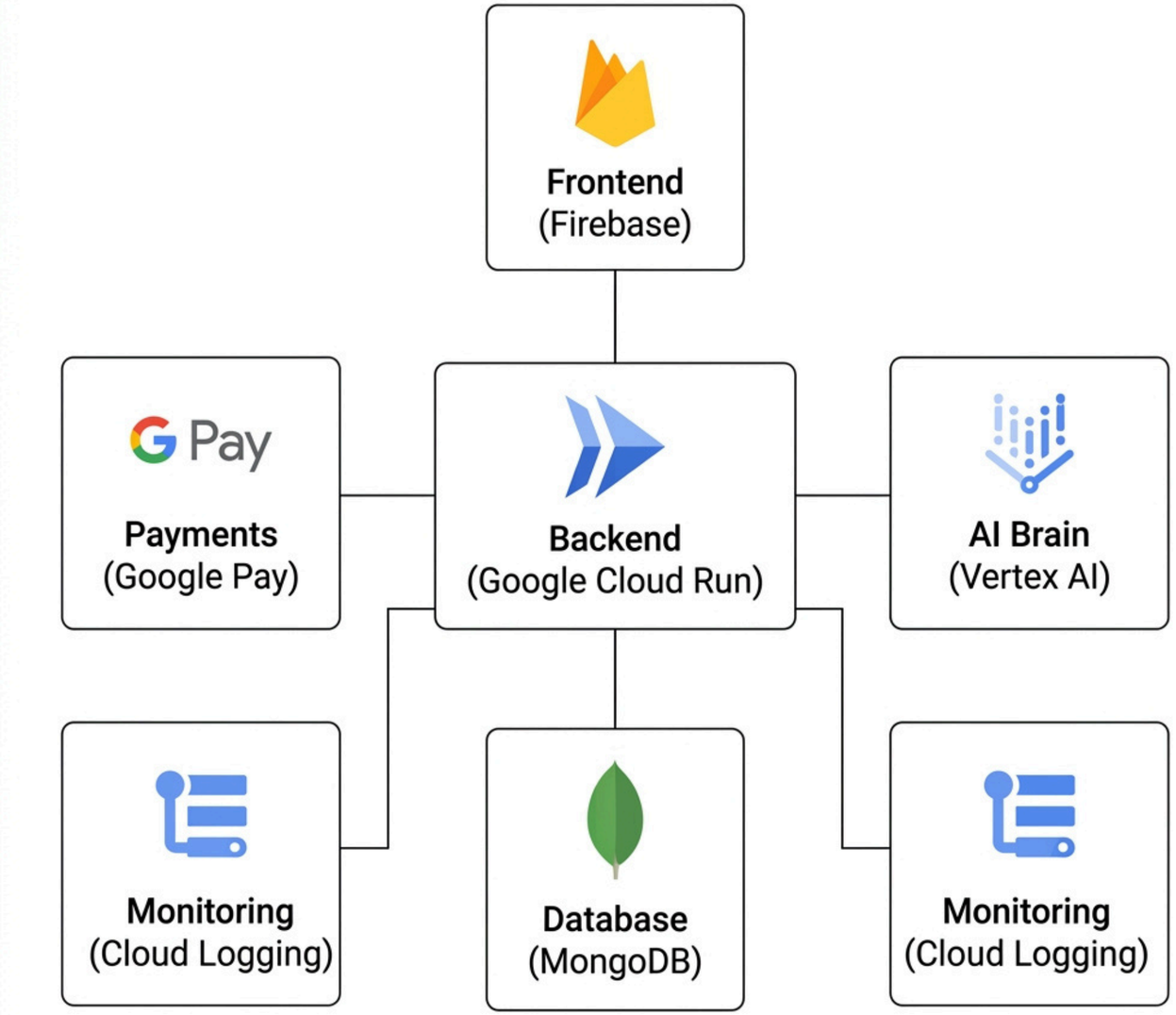
List of features offered by Docvail

- **AI Symptom Analyzer:** Manages complex, unstructured symptom descriptions into precise medical specialties using Gemini.
- **Dynamic Real-Time Scheduling:** Patients view live availability of doctors and can book instant, confirmed time slots.
- **Doctor Confortability:** Can easily update the status of availablilty on the portal , No overcrowding , No long ques.
- **Patient Comfortability:** Can easily compare among the wide options of the Docotrs and can select anyone.
- **Secure Authentication:** High-security, Dual-Identifier (Email/Mobile) login flow with OTP verification for both patients and providers.

Use-case diagram



Architecture diagram



- **Google Technologies:** Google Vertex AI(Gemini 2.5), Google Cloud Run(Compute), Google Firebase Hosting (CDN), Google Pay API(Payments), and Google Cloud Logging.
- **Frontend:** React, TailwindCSS, Axios.
- **Backend:** Node.js, Express.js.
- **Database:** MongoDB
- **Payment Gateway:** Razorpay (UPI(G pay), NetBanking, and Card processing)
- **Security:** JSON Web Tokens (JWT), Bcrypt.

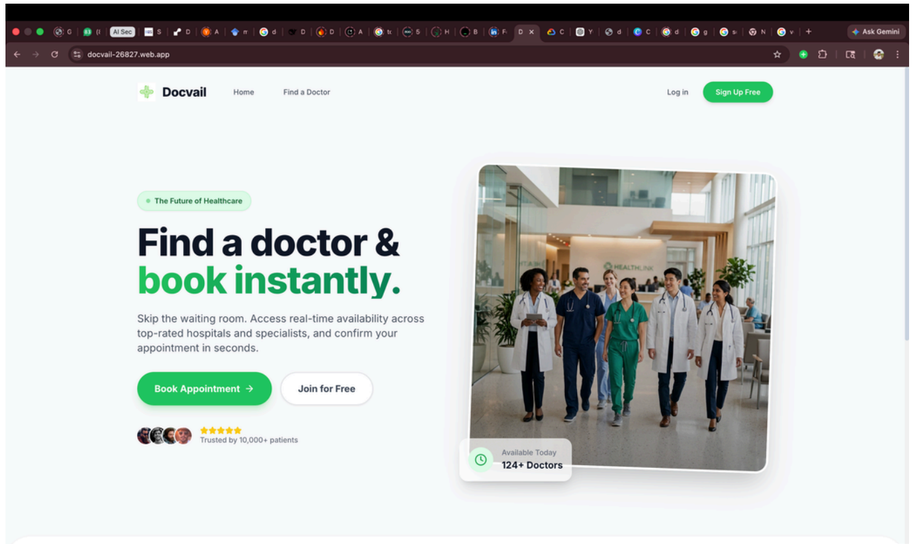
Estimated implementation cost

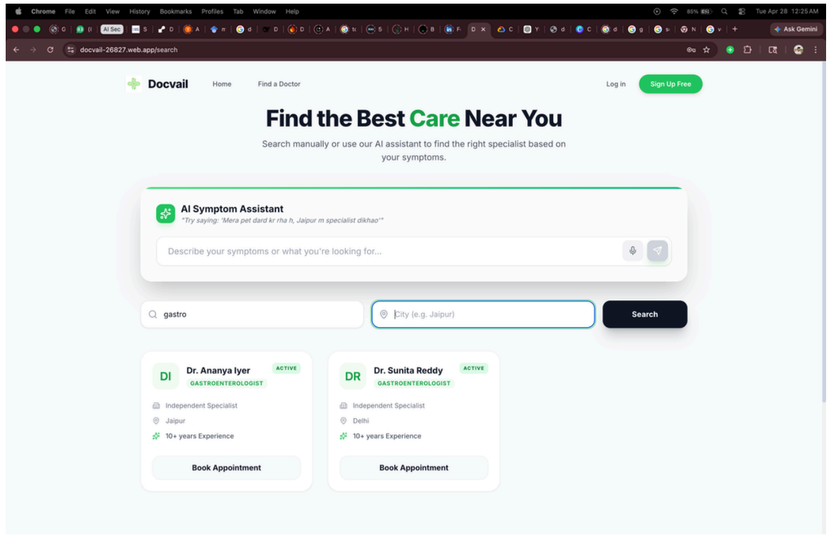
Phase 1: Prototype Status (Current) Total Monthly Cost: ₹0
Leveraging Google for Startups credits and generous Free Tiers for Vertex AI, Cloud Run, and Firebase Spark Plan.

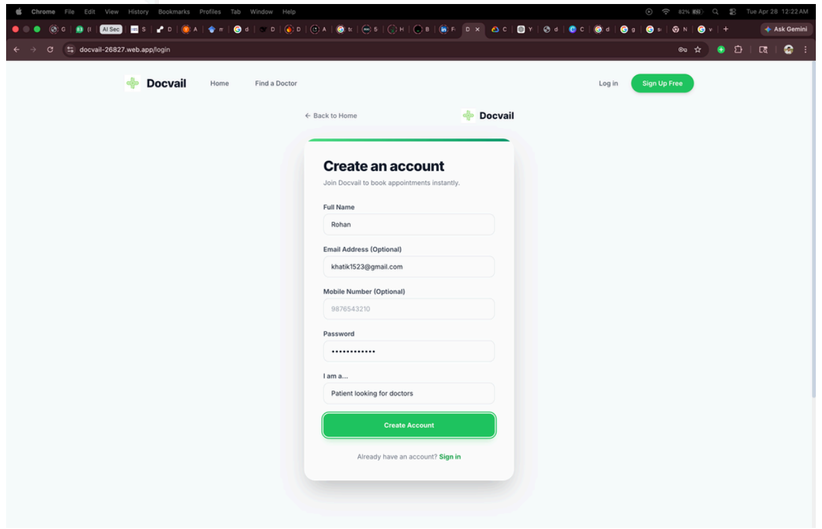
Phase 2: Pilot Implementation (Target: 1 District / 15+ Clinics)
Cloud Infrastructure Setup: ~₹15,000 (Vertex AI & Cloud Run production scaling).
Hospital Digitalization:~₹75,000 (One-time setup for clinic dashboards & staff training).
Regulatory Compliance:~₹40,000 (ABHA/ABDM integration & security audits).
Local Patient Awareness: ~₹50,000 (On-ground banners & localized SMS outreach).
Total Setup Budget:~₹1.80 Lakh (Pilot Ready)**

Scalability & Revenue: Revenue Model: 10% commission per successful booking.
Expansion :Once the ₹1.80L setup is proven in one district, the model can be replicated across multiple states with high capital efficiency.

Snapshots of the MVP







Future Development

- Hybrid Payment Ecosystem: Full deployment of Google Pay API for digital users and "Pay at Counter" offline support for rural accessibility.
- Regional Language Voice Support: Utilizing Google Speech-to-Text for 10+ Indian languages.
- Telemedicine Integration: Adding secure WebRTC video consultations.
- Google Maps Integration: Embedding Maps API for clinic navigation and live traffic.

Provide links to your:

1. GitHub Public Repository: https://github.com/as3121945-hash/Docvail
2. Demo Video Link (3 Minutes):
3. Working Prototype Link: https://docvail-26827.web.app/

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Thank You